KPK BOARD TEXTBOOK NOTES

Physics - Class 9

Unit 9: Transfer of Heat

Complete notes in English, based on *A Textbook of Physics, Grade IX*
Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar

> **How to use these notes.** Read one section at a time, learn the green definition boxes by heart, and copy every yellow formula box into your notebook. Work through the solved examples and numericals before you try the multiple-choice questions (MCQs) at the end. The last page is a one-page revision sheet for the night before the exam.

What this unit covers

1. Heat and the three ways it travels - conduction, convection and radiation
2. Conduction of heat - its mechanism, good and bad conductors, and its uses
3. Thermal conductivity - the rate-of-flow formula and Table 9.1 values
4. Convection of heat - convection currents and their everyday applications
5. Radiation of heat - good and bad emitters and absorbers, and the thermos flask
6. The greenhouse effect and global warming
7. The book's exercise: MCQs with an answer key, and solved numericals

After studying this unit you should be able to (book checklist)

- Recall that thermal energy is transferred from a region of higher temperature to a region of lower temperature.
- Describe, in terms of molecules and electrons, how heat transfer occurs in solids.
- State the factors affecting the transfer of heat through solid conductors and so define thermal conductivity.
- Solve problems based on the thermal conductivity of solid conductors.
- Give examples of good and bad conductors of heat and describe their uses.
- Explain convection currents in fluids due to difference in density, with everyday examples.
- Explain how insulation reduces energy transfer by conduction.
- Describe radiation from all objects, and explain that it needs no material medium; its rate depends on the colour and texture of the surface, the surface temperature and the surface area.

Heat - Energy on the Move

> **Heat (Q)** is the thermal energy that can be transferred between two systems because of a temperature difference. When two bodies are at different temperatures, thermal energy transfers from the one with the **higher** temperature to the one with the **lower** temperature.

Heat transfer has an enormous range of applications. Our survival and comfort depend on keeping our bodies at a constant temperature, and keeping a house at a comfortable temperature in winter and in summer - economically and efficiently - requires a knowledge of how heat travels.

There are **three modes** of heat transfer. A campfire shows all three at once:

Mode	How heat travels	Needs a material medium?
Conduction	Particle to particle, through matter, without movement of the matter as a whole	Yes - matter itself
Convection	By the bulk (actual) movement of a heated fluid	Yes - a liquid or a gas
Radiation	As electromagnetic waves (heat rays)	No - it works even through empty space

9.1 Conduction of Heat

The handle of a metal spoon held in hot tea soon gets warm. The heat passes along the spoon by conduction.

> **Conduction** is the flow of thermal energy (heat) through matter from places of higher temperature to places of lower temperature, **without movement of the matter as a whole**.

9.1.1 Mechanism of heat conduction

Heat one end of an iron bar in a burner flame. After a few seconds the other end becomes too hot to hold. How does the heat travel?

- A solid is made of closely packed atoms. The greater the temperature, the greater the kinetic energy of its atoms or molecules.
- Atoms in the hotter part vibrate more violently than those in the colder part.
- These wildly vibrating atoms collide with their neighbouring atoms and transfer some of their energy to them, and those pass energy to their own neighbours, and so on.
- Metals also contain a large number of **free electrons** - electrons free to move through the metal. They gain kinetic energy at the hot end, diffuse into the colder part, and transfer energy by collisions there. This is why metals conduct so well.

> **In short:** conduction is the process by which heat energy is transferred **from particle to particle**.

9.1.2 Good and Bad Thermal Conductors

> **Good conductors (heat conductors)** allow heat to pass through them easily - for example copper, aluminium and iron. **Bad conductors (heat insulators)** do not allow heat to flow through them easily - for example wood, cork, wool, plastic foam and air.

Activity 9.1 - Comparing the conducting power of metals

Matchsticks are fixed with melted wax to one end of rods of copper, aluminium, brass and iron. The other ends are heated together by a burner. As each far end reaches the melting point of the wax, its matchstick drops off. The matchstick on **copper falls first** - copper is the best conductor - followed by aluminium, then brass, and then iron.

9.1.3 Practical Applications of Conduction

A **Cooking pots and pans** are made of metals, which are good conductors. They conduct heat readily to the food inside and spread it evenly.

B **Plastic foam and fibreglass** insulators are used in the walls and ceilings of homes to keep them cool in summer and warm in winter. They insulate well because they contain tiny pockets of trapped air.

C **Wire gauze** placed over a flame conducts heat outwards from the flame, so a glass beaker can be heated safely on it, protected from the concentrated heat.

D **Pot holders and table mats** for hot pans are made of poor conductors such as cloth and wood, so our hands do not burn.

E **Woollen clothes** have fine pores filled with air. Air and wool are both bad conductors, so body heat does not flow out and the body stays warm in winter.

F **Igloos** - shelters built of ice and snow - work because ice and snow act as thermal insulation and minimise the loss of heat from inside by conduction.

G **Ice box** has a double wall of tin or iron, and the space between the walls is filled with cork, a poor conductor. The cork prevents outside heat flowing in, so the ice does not melt.

H **Soldering:** when objects are in direct contact - such as the soldering iron and the circuit board - heat is transferred by conduction.

9.2 Thermal Conductivity

Thermal conductivity of a substance is a measure of the ability of the substance to conduct heat energy.

Consider a rod (slab) of length *L* and cross-sectional area *A*. Its hot face is at temperature T_h and its colder face at T_c, with $T_h > T_c$. The rate of flow of heat Q/t across the slab depends on four factors:

1. **Temperature difference** ($T_h - T_c$) between the two faces - the greater the difference, the greater the heat transferred per unit time.
2. **Length of the rod L** - the greater the length, the smaller the rate of flow of heat through it.
3. **Cross-sectional area A** - a larger area allows a greater rate of heat flow.
4. **The nature (material) of the slab.**

$$Q / t = k \times A \times (T_h - T_c) / L \quad \text{--- (9.1)}$$

Here *k* is the **thermal conductivity** of the material. Rearranging:

$$k = Q \times L / [A \times t \times (T_h - T_c)] \quad \text{--- (9.2)}$$

Definition of thermal conductivity

As a special case, put L = 1 m, A = 1 m^2, $T_h - T_c$ = 1 K (or 1 °C) and t = 1 s in equation (9.2). Then k = Q, so:

Thermal conductivity k of a substance is the quantity of heat that flows through one square metre of area of the substance in one second when a temperature difference of one kelvin is maintained across a thickness of one metre.

Unit: W m^{-1} K^{-1}, which is the same as J m^{-1} K^{-1} s^{-1}.

Reading the formula. Heat flows faster when k, A or the temperature difference is large, and slower when the slab is thick (large L). To insulate, choose a material with a very small k and make it thick - exactly what wool, cork and foam do.

Table 9.1 - Thermal Conductivities of Common Substances

Substance	k (J m^{-1} K^{-1} s^{-1})	Substance	k (J m^{-1} K^{-1} s^{-1})
Silver	420	Steel (stainless)	14
Copper	390	Ice	2.2
Gold	318	Concrete brick	0.84
Aluminium	220	Glass (average)	0.8
Brass	105	Water	0.6
Steel iron	80	Wood	0.08 - 0.16
Lead	35	Cotton	0.08
Cork	0.042	Wool	0.04
Plastic foam	0.033	Air	0.023
Styrofoam	0.010		

Metals have large k values (good conductors); materials with tiny k values - cork, wool, foam, air, Styrofoam - are the insulators used in clothing, ice boxes and buildings.

Example 9.1 (book) - Styrofoam ice box

A Styrofoam (k = 0.010 W m^{-1} K^{-1}) ice box has total area A = 0.950 m^2 and walls of average thickness L = 2.50 cm = 0.0250 m. Inside, ice and drinks are at 0 °C; outside is 35.0 °C. Find the rate of energy transfer.

Temperature difference = 35 - 0 = 35 K.

$$Q/t = 0.010 \times 0.950 \times 35 / 0.0250 = \mathbf{13.3\ W} = \mathbf{13.3\ J\ s^{-1}}$$

Assignment 9.1 (book) - solved: concrete wall

Find the heat transferred in one hour through a concrete wall of area 6.9 m^2, thickness 0.20 m, with one side at 20 °C and the other at 5 °C (k for concrete = 1.3 J m^{-1} K^{-1} s^{-1}).

Q/t = 1.3 × 6.9 × (20 - 5) / 0.20 = 672.75 W. In one hour (3600 s):

$$Q = 672.75 \times 3600 = \mathbf{2.42 \times 10^6\ J}$$

9.3 Convection of Heat

We can warm our hands by holding them over an open flame: the air directly above the flame is heated and expands, and heat is transferred by the movement of the heated particles.

> **Convection** is the transfer of heat from one place to another by the **bulk motion of fluids**. Convection occurs only in fluids (liquids and gases). It cannot occur in solids, because the atoms in a solid are in fixed positions and cannot change place. For the same reason, convection occurs very easily in gases.

Activity 9.3 - Seeing convection currents

Drop a few crystals of potassium permanganate to the bottom of a beaker of water and heat the beaker gently just below the crystals. Purple streaks of coloured water rise, and cold water from the sides takes their place. The water keeps circulating and grows hotter and hotter - each part is heated in turn. The currents set up in this process are called **convection currents**.

9.3.1 Mechanism of heat convection

1. The portion of the fluid at the bottom is heated first.
2. It expands and becomes **less dense**.
3. Being less dense, the warm fluid **moves upward**.
4. It is replaced by the colder, denser fluid around it, which flows down to the point of heating.
5. That fluid in its turn absorbs heat, expands and is pushed upward - so a continuous circulatory flow is established from the bottom to the top.

> **Convection currents** are the currents set up in a fluid when heat is transferred by convection.

Everyday sign: black marks often appear on the wall or ceiling above a lamp. They are caused by dust carried upwards in the air convection currents produced by the hot lamp or radiator.

Point to ponder (book): you can hold your fingers beside a candle flame without harm, but not above it. Hot air travels upward by convection, and air is a poor conductor, so very little heat travels sideways to your fingers.

9.3.2 Practical Applications of Heat Convection

A **Heating water.** Without convection currents it would be very difficult to boil water. In an electric kettle the lower layers are warmed first; the heated water expands, its density falls, and it moves up to the top. Dense cool water replaces it at the bottom and is heated in turn.

B **Sea breeze and land breeze.** Convection causes coastal breezes. By day the land heats up more quickly than the sea; hot air over the land rises and cold air from the sea blows in to replace it - a **sea breeze**. At night the land cools more quickly than the sea; hot air over the sea rises and cold air from the land blows out to replace it - a **land breeze**.

C **Riding on thermals.** Thermals are streams of hot air rising in the sun - they are convection currents. Birds can fly for hours on thermals without flapping their wings, and glider aeroplanes rise by riding on thermals.

D **Refrigerator.** Convection circulates cold air around the food. Air is cooled by the freezer compartment at the **top** of the refrigerator; as it sinks, warmer air rises from below to replace it. The circulating air carries heat energy away from all the food in the fridge.

E **Ventilation.** Ventilators are fitted near the ceiling of classrooms and houses. Warm, stale air inside the room rises and escapes through them, and fresh cold air is drawn in through doors and windows. In the same way, smoke and hot gases from fires in houses and factories rise and escape through chimneys.

> **Tid-bit (book).** A burning candle heats the air near its flame. Hot air is less dense and more buoyant than cool air, so a circulation is set up: hot air rises and cool, oxygen-rich air replaces it from below. That convection is necessary for the candle to keep burning.

Conduction or convection? A quick test

Question	Conduction	Convection
Does the material itself move?	No	Yes - the fluid circulates
Works in solids?	Yes - best in metals	No
Works in liquids and gases?	Slowly (they are poor conductors)	Yes - this is their main mode

9.4 Radiation of Heat

Sit next to an open campfire. The heat does not reach you by conduction, because air is a bad conductor. It does not reach you by convection, because hot air rises up. It reaches you by **radiation**: the hot fire radiates heat rays, just as it radiates light rays, and when these rays fall on anything they make that object hotter.

The Sun's heat reaches us after passing through millions of kilometres of empty space. Neither conduction nor convection can take place in a vacuum, because no material is present. Sun heat reaches us by radiation of heat.

> **Radiation of heat** is the heat transfer from a hotter place to a colder place **with or without a material medium in between**. Unlike conduction and convection, radiation does not necessarily require a material medium.

9.4.1 Mechanism of radiation of heat

- Radiation is energy transfer by **electromagnetic waves**. Electromagnetic radiation comes from accelerating electric charges.
- On a molecular level, as objects warm up their molecules vibrate harder and harder, accelerating electric charges - and those charges emit radiation.
- Heat radiation is like light, except that it is not visible, or barely visible: the heat part lies in the **infrared** part of the electromagnetic spectrum.
- Electromagnetic waves can transfer energy through a vacuum or empty space, as well as through a material medium such as glass.
- **Every object radiates continually**, unless its temperature is at absolute zero (0 K). Even a scoop of ice cream at about 237 K radiates, and our own bodies radiate all the time - in infrared, which is why infrared scopes can "see" people in the dark.

> **Radiant energy** is the energy emitted from a hotter place and carried through radiation.

Three modes compared

	Conduction	Convection	Radiation
Medium needed	Yes	Yes (fluid)	No
Particles move?	No - they only vibrate	Yes - bulk motion	No particles needed
Example	Spoon in hot tea	Sea breeze	Sunlight warming Earth

9.4.2 Good and Bad Emitters and Absorbers

- Some surfaces absorb radiation better than others: a **black** surface absorbs most, and a **white** surface reflects most. A dull black kettle absorbs heat better than a silver kettle, and a black car standing in the sun warms up more quickly than a car of any other colour.
- Some surfaces also emit radiation better: a black saucepan cools down more quickly than any other, and hot water in a kettle covered with soot cools faster than in a similar kettle with a shining surface.
- **Activity 9.4:** hold the backs of your hands on either side of a hot copper sheet, one side polished and the other blackened. The dull black side feels hotter - it is the better emitter.

> **Rule:** a good absorber is a good radiator (emitter), and a poor absorber is a poor radiator.

Rate of energy transfer by radiation depends on

1. Colour and texture of the surface
2. Surface temperature
3. Surface area

That is why **white clothes** are worn in hot climates (white is a good reflector and a poor absorber) and **black or dark clothes** in cold climates (dark is a poor reflector and a good absorber).

9.4.3 Practical Applications of Radiation

A **Colouring materials.** The cooling fins on the heat exchanger at the back of a refrigerator are painted **black** so they lose heat more quickly. By contrast, polished saucepans are poor emitters and keep their heat longer.

B **Texture of the surface.** Radiant-barrier material such as ARMA foil is used to reflect radiant heat.

C **Satellite protective coating.** The highly reflective metal foil covering a satellite (the book shows the Hubble Space Telescope) minimises temperature changes.

D **Thermos flask** - the vessel designed to prevent heat transfer by all three modes. It is a double-walled glass vessel, silvered on the inside:

- **Silvering** reflects all radiant heat trying to enter or leave - it stops radiation.
- The space between the walls is highly evacuated (**vacuum**) - this prevents convection (and conduction through air).
- **Glass** is a poor conductor, so it minimises conduction.

Heat loss is so small that a hot liquid stays hot for a very long time, and a cold thing placed in the flask stays cold just as long.

9.5 Greenhouse Effect and Global Warming

> **Greenhouse effect** is the process by which radiation from a planet's atmosphere warms the planet's surface to a temperature above what it would be without its atmosphere. Its strength depends on the atmosphere's temperature and on the amount of greenhouse gases the atmosphere contains.

9.5.1 Mechanism of the greenhouse effect

1. Earth receives energy from the Sun as ultraviolet, visible and near-infrared radiation.
2. Of the solar energy at the top of the atmosphere, about **26% is reflected to space** by the atmosphere and clouds, and about **19% is absorbed** by the atmosphere and clouds. Most of the remaining energy - more than 50% - is absorbed at Earth's surface.
3. Earth's surface is much colder than the Sun, so it radiates back at much **longer wavelengths** than those it absorbed.
4. Most of this outgoing thermal radiation is **absorbed by the atmosphere**, warming it. The atmosphere also gains heat by latent heat fluxes from the surface.
5. The atmosphere radiates energy both upwards and downwards. The downward part is absorbed by Earth's surface, giving a higher equilibrium temperature than if the atmosphere were absent.

9.5.2 Importance - and the global warming concern

- The greenhouse effect is essential for life. Natural greenhouse gases - water vapour, carbon dioxide (CO_2), methane (CH_4) and ozone (O_3) - keep the air near Earth's surface about **33 °C warmer** than it would otherwise be. Without the atmosphere, Earth's average temperature would be well below the freezing point of water.
- Human activity has increased greenhouse gases, causing **global warming** - an increase in Earth's temperature. The book records that from 1880 to 2012 the global average temperature increased by **0.85 °C**.
- The largest human influence has been carbon dioxide emitted from factories and motor vehicles, by burning fossil fuels. About half of the CO_2 released is not absorbed by vegetation and the oceans and remains in the atmosphere.
- To decrease global warming we must reduce the emission of greenhouse gases and plant more vegetation to absorb the carbon dioxide produced.

Note (book): a real greenhouse - and a car parked in the hot sun with its windows closed - warms mainly by *reducing convection* of air, while the atmospheric greenhouse effect works by preventing absorbed heat from leaving through radiative transfer. The name is an analogy, not the same mechanism.

Exercise MCQs (Book) - With Answer Key

The book's exercise prints nine questions (the number 5 is printed twice); they are numbered 1-9 here in book order. Answers were checked against the unit text and Table 9.1.

1. Which of the following is the best heat conductor? A. aluminium B. tin C. iron D. copper
2. Identical cubes of the following materials are kept in a room at the same temperature. Which will feel coldest by touching them? A. Wood B. Glass C. Iron D. Styrofoam
3. The transfer of heat by convection is smallest in: A. solids B. liquids C. gases D. none
4. Heat is transferred inside the human body by the flow of blood. Which process best describes this action of the blood? A. convection B. conduction C. radiation D. none
5. The best absorber of radiation is a body whose surface is: A. White B. Grey C. Dull black D. Highly polished
6. The mode of transfer of heat which does not require a material medium is called: A. convection B. conduction C. radiation D. none
7. Which of the following has the highest thermal conductivity? A. water B. wood C. wool D. air
8. The temperature at which a body is not radiating any heat is: A. 0 °C B. 0 °F C. 0 K D. all of these
9. The electrons that are free to move through the metal are called: A. lose electrons B. free electrons C. conduction electrons D. holes

Answer key: 1-D 2-C 3-A 4-A 5-C 6-C 7-A 8-C 9-B

Why: copper (390) beats aluminium, iron and tin in Table 9.1; iron feels coldest because it conducts heat away from the hand fastest; convection cannot happen in solids; moving blood is a moving fluid, so convection; dull black absorbs most; radiation needs no medium; water (0.6) beats wood, wool and air; a body stops radiating only at 0 K.

Conceptual questions (book) - one-line answers

- *Why are white clothes preferred in summer?* White reflects most radiation and absorbs least, so we stay cooler.
- *Why is the freezer at the top of a refrigerator?* Cold air sinks and sets up convection currents that cool the whole fridge.
- *Why does a black car in the sun warm up fastest?* Black is the best absorber of radiation.
- *Why does a tile floor feel colder than a carpeted floor?* Tile conducts heat away from bare feet faster; carpet is an insulator.
- *How do woollen sweaters keep us warm?* Wool and its trapped air are bad conductors, so body heat cannot escape easily.
- *How can birds fly for hours without flapping?* They ride rising convection currents - thermals.

Solved Numericals (Book Exercise)

Numerical 1 - Heat loss through woollen clothing

A person's body is covered with 1.6 m^2 of wool clothing of thickness 2.0×10^{-3} m. The outside surface of the wool is at 11 °C and the skin temperature is 36 °C. k for wool = 0.04 W m^{-1} K^{-1}. How much heat per second is lost by conduction?

$$Q/t = 0.04 \times 1.6 \times (36 - 11) / 0.002 = \mathbf{800\ W}$$

Numerical 2 - Heat loss through a brick wall

The external wall of a brick house has area 16 m^2 and thickness 0.3 m. Inside temperature is 20 °C and outside 0 °C. k for brick = 0.84 W m^{-1} K^{-1}. Calculate the rate of heat loss through the wall.

$$Q/t = 0.84 \times 16 \times (20 - 0) / 0.3 = \mathbf{896\ W}$$

Numerical 3 - Heat through a window glass

Window glass has k = 0.8 W m^{-1} K^{-1}. At what rate is heat conducted through a window of area 2.0 m^2 and thickness 4.0 mm, if the inside of the air-conditioned room is 20 °C and the outdoors temperature is 35 °C?

$$Q/t = 0.8 \times 2.0 \times (35 - 20) / 0.004 = \mathbf{6000\ W} = \mathbf{6.0\ kW}\ \text{(flowing into the room)}$$

Comprehensive questions (book) - for practice

1. Explain conduction of heat and its mechanism. Describe any three of its practical applications.
2. Explain thermal conductivity of a substance and its mathematical description.
3. Explain convection of heat and its mechanism. Explain any three of its practical applications.
4. Explain radiation of heat and its mechanism. Describe any three of its practical applications.
5. Discuss the greenhouse effect. Explain its importance and the global warming concern.

Every section of these notes answers one of these questions - use the green definition boxes as the core of each answer.

One-Page Revision - Unit 9

Key points (book)

- **Heat transfer:** the process in which heat travels from one place to another because of a difference of temperature.
- **Conduction of heat:** the particle-to-particle mode of heat transfer by collisions or indirect interaction.
- **Thermal conductivity:** a measure of the ability of a substance to conduct heat energy.
- **Convection of heat:** transfer of heat by the actual motion of the heated substance; the currents set up are convection currents.
- **Radiation of heat:** heat and energy transfer from a hotter place to a colder place with or without a material medium in between. The energy carried is radiant energy.

Formulas to remember

$$Q/t = k\,A\,(T_h - T_c)\,/\,L \quad | \quad k = Q\,L\,/\,[\,A\,t\,(T_h - T_c)\,] \quad | \quad \text{unit of } k\text{: W m}^{-1}\text{ K}^{-1}$$

Numbers to remember

- k values: silver 420, copper 390, aluminium 220 - and insulators: cork 0.042, wool 0.04, air 0.023, Styrofoam 0.010.
- Greenhouse numbers (book): 26% of solar energy reflected to space, 19% absorbed by atmosphere and clouds, more than 50% absorbed at Earth's surface; natural greenhouse effect keeps Earth about 33 °C warmer; global average rose 0.85 °C from 1880 to 2012.
- Example 9.1 answer: 13.3 W. Exercise numericals: 800 W, 896 W, 6000 W.

Exam traps

- Convection happens **only in fluids** - never in solids.
- Radiation is the **only** mode that needs no material medium.
- A good absorber is also a good emitter; dull black is best at both, shiny surfaces are worst.
- In the conduction formula, thickness L goes on the **bottom**: thicker slab, slower heat flow.
- Convert thickness to metres (cm and mm) before using the formula, and remember a temperature difference in °C equals the same difference in K.
- The freezer sits at the **top** of a refrigerator so that sinking cold air sets up convection.

> **Source:** *A Textbook of Physics, Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar - Unit 9, Transfer of Heat (book pages 233-251). Notes prepared in English for Class 9 students.